Introduction to Advanced CSS

HTML stands for HyperText Markup Language. It is used to design web pages using a markup language. HTML is the combination of Hypertext and Markup language. Hypertext defines the link between the web pages. A markup language is used to define the text document within a tag which defines the structure of web pages. This language is used to annotate (make notes for the computer) text so that a machine can understand it and manipulate text accordingly. Most markup languages (e.g. HTML) are human-readable. The language uses tags to define what manipulation has to be done on the text.

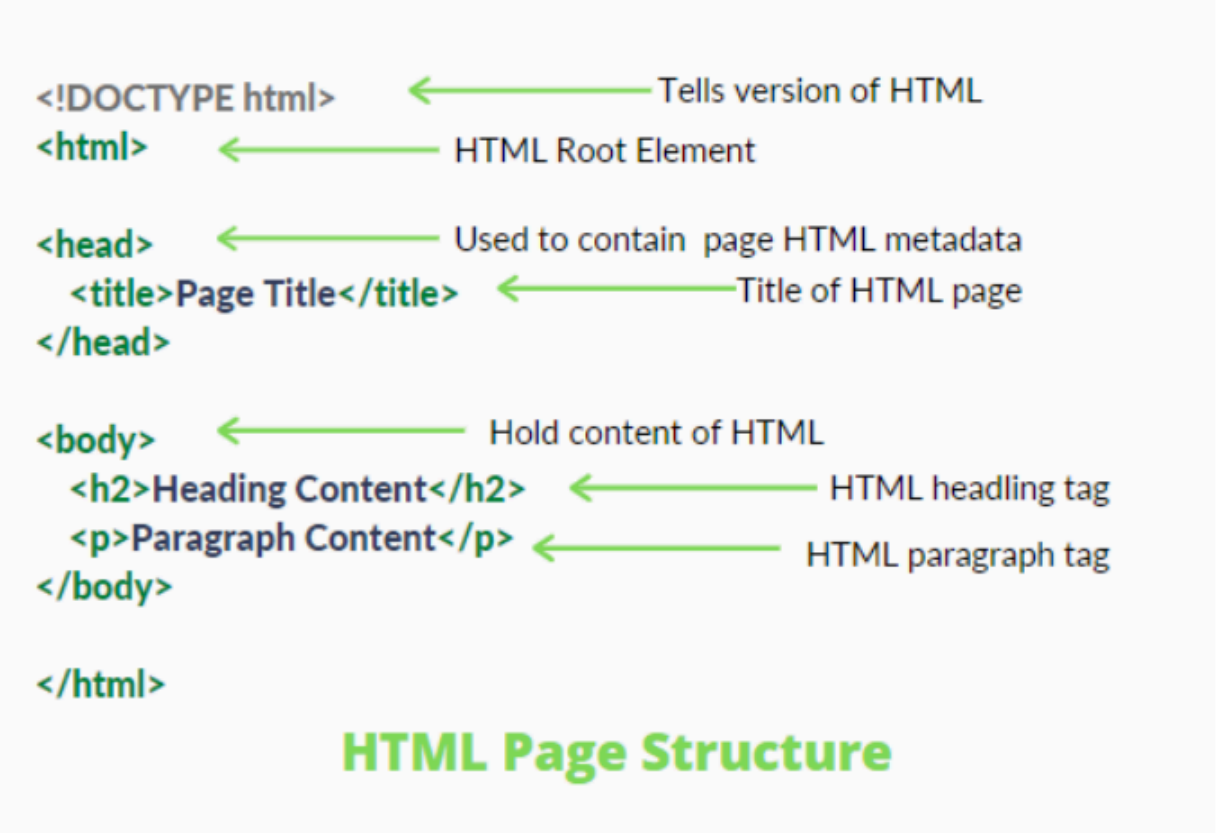
HTML is a markup language used by the browser to manipulate text, images, and other content, in order to display it in the required format. HTML was created by Tim Berners-Lee in 1991. The first-ever version of HTML was HTML 1.0, but the first standard version was HTML 2.0, published in 1999.

Elements and Tags

HTML uses predefined tags and elements which tell the browser how to properly display the content. Remember to include closing tags. If omitted, the browser applies the effect of the opening tag until the end of the page.

HTML page structure

The basic structure of an HTML page is laid out below. It contains the essential building-block elements (i.e. doctype declaration, HTML, head, title, and body elements) upon which all web pages are created.



Important Tags

<DOCTYPE! html>:
This is the document type declaration (not technically a tag). It declares a document as being an HTML document. The doctype declaration is not case-sensitive.

<html>
This is called the HTML root element. All other elements are contained within it.

<head>
The head tag contains the “behind the scenes” elements for a webpage. Elements within the head aren’t visible on the front-end of a webpage. HTML elements used inside the <head> element include

<title>
The <title> tag in HTML is used to define the title of an HTML document. It sets the title in the browser toolbar. It provides the title for the web page when it is added to favorites.

<body>
The body tag is used to enclose all the visible content of a webpage. In other words, the body content is what the browser will show on the front-end.

<a>
Anchor tag defines hypertexts. (Used to link another website)

<h1 .... h6>
It is Used to define headings from level 1 to level 6.

<img>
Used to insert images into a website.

** **
These tags help us to write paragraph statements on a webpage. They start with the tag and end with . Here the <br> tag is used to break line and acts as a carriage return.
 is an empty tag.

**<b> or **

Use to style text into bold format.

<I>

Used to style text into Italic format.

Introduction to CSS

CSS stands for Cascading Style Sheets. It is the language for describing the presentation of Web pages, including colours, layout, and fonts, thus making our web pages presentable to the users.

CSS is designed to make style sheets for the web. It is independent of HTML and can be used with any XML-based markup language. Now let's try to break the acronym:

- Cascading: Falling of Styles
- Style: Adding designs/Styling our HTML tags
- Sheets: Writing our style in different documents

How can CSS be added?

CSS is added to HTML pages to format the document according to information in the style sheet. There are three ways to insert CSS in HTML documents.

1. Inline CSS
2. Internal CSS
3. External CSS

1) Inline CSS

Inline CSS is used to apply CSS on a single line or element.

For example:

```
<p style="color:blue">Hello CSS</p>
```

2) Internal CSS

Internal CSS is used to apply CSS on a single document or page. It can affect all the elements of the page. It is written inside the style tag within the head section of html.

For example:

```
 <style>

p{color:blue}

</style>
```

3. External CSS

External CSS is used to apply CSS on multiple pages or all pages. Here, we write all the CSS code in a css file. Its extension must be .css for example style.css.

For example:

```
p{color:blue}
```

You need to link this style.css file to your html pages like this.

```
<link rel="stylesheet" type="text/css" href="style.css">
```

The link tag must be used inside the head section of html.

CSS Selectors

CSS selectors are used to "find" (or select) the HTML elements you want to style.

We can divide CSS selectors into five categories:

- Element selectors
- Id Selector
- class selectors
- Universal selector

1. The CSS element Selector

The element selector selects HTML elements based on the element name.

Example:
Here, all <p> elements on the page will be center-aligned, with a red text color:

```
p {
  text-align: center;
  color: red;
}
```

2. The CSS id Selector

The id selector uses the id attribute of an HTML element to select a specific element.

The id of an element is unique within a page, so the id selector is used to select one unique element!

To select an element with a specific id, write a hash (#) character, followed by the id of the element.

Example:

The CSS rule below will be applied to the HTML element with id="para1":

```
#para1 {
  text-align: center;
```

```
  color: red;
}
```

3. The CSS class Selector

The class selector selects HTML elements with a specific class attribute.

To select elements with a specific class, write a period (.) character, followed by the class name.

Example:

In this example all HTML elements with class="center" will be red and center-aligned:

```
.center {
 text-align: center;
  color: red;
}
```

4. The CSS Universal Selector

The universal selector (*) selects all HTML elements on the page.

Example:

The CSS rule below will affect every HTML element on the page:

```
* {
  text-align: center;
  color: blue;
}
```

1. Design a dynamic web page to build a responsive resume?

Resume.css

```
.heading{
  display:flex;
  justify-content:center;
}
.intro{
  display:flex;
  flex-direction:row;
  align-items: flex-start;
}
.image{
  display:flex;
```

```
  justify-content: center;
  align-items: center;
  margin-left: 20px;
}
.image img{
   width: 200px;
   height: 200px;
   border-radius: 50%;
}
body{
  font-family:serif;
}
#para{
  display:flex;
  height:25vh;
}
```

Resume.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Resume</title>
    <link rel="stylesheet" href="Resume.css">
</head>
<body>
    <div class="heading"><h2><b>RESUME</b></h2></div>
    <h3>Introduction</h3>
    <div class="intro">
        <p id="para">I am Penkey Prajna Sree. I am currently studying III year of B-TECH in Vishnu Institute of Technology,Bhimavaram in the department of Information Technology.I have looking for some internships to have some experiences and gain some skills in python and AI Technologies.
            Lorem ipsum dolor sit amet consectetur, adipisicing elit. Consequatur dolorum nam est voluptatibus corporis accusamus labore totam ex? Exercitationem, dicta fugit! Perspiciatis nesciunt, fuga perferendis sit minima voluptate inventore incidunt.
            Lorem ipsum dolor sit amet consectetur adipiscing elit. Numquam quod, cupiditate repudiandae necessitatibus animi suscipit! Laborum ratione aspernatur debitis eum veniam distinctio provident accusamus quod quidem dolore, eos numquam sit?
            Lorem ipsum dolor sit, amet consectetur adipisicing elit. Iste voluptates debitis a dolorem nihil neque perferendis modi officiis! Aperiam doloremque dolores ducimus explicabo possimus alias aspernatur tenetur praesentium mollitia quam.
            Lorem ipsum dolor sit amet consectetur adipiscing elit. Inventore necessitatibus natus, recusandae incidunt quasi numquam enim id corrupti iure adipisci. Sed nesciunt eum quod autem, totem debitis at soluta cupiditate?
            Lorem ipsum dolor sit amet consectetur adipiscing elit. Eligendi quo incidunt deleniti asperiores animi voluptatem ab reprehenderit laborum illum consequuntur excepturi, placeat, impedit rem doloremque quia explicabo. Ullam, excepturi odio!
```

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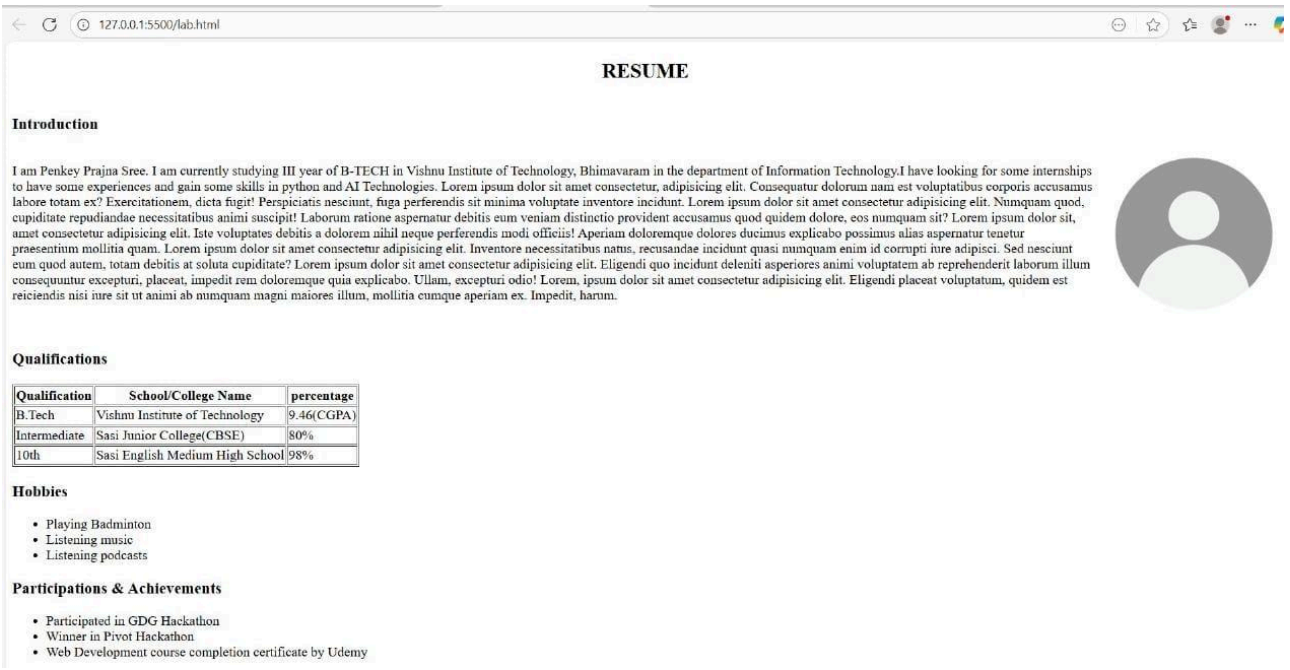
```
        </p>
            <div class="image"><img
src="https://www.shutterstock.com/image-vector/user-profile-icon-vector-avatar-600nw-2558760599.jpg"></div>
    </div>
    <h3 id="Q">Qualifications</h3>
    <table border="1">
        <tr>
            <th>Qualification</th>
            <th>School/College Name</th>
            <th>percentage</th>
        </tr>
         <tr>
            <td>B.Tech</td>
            <td>Vishnu Institute of Technology</td>
            <td>9.46(CGPA)</td>
        </tr>
        <tr>
            <td>Intermediate</td>
            <td>Sasi Junior College(CBSE)</td>
            <td>80%</td>
        </tr>
        <tr>
            <td>10th</td>
            <td>Sasi English Medium High School</td>
            <td>98%</td>
        </tr>
    </table>
    <h3>Hobbies</h3>
    <ul>
        <li>Playing Badminton</li>
        <li>Listening music</li>
        <li>Listening podcasts</li>
    </ul>
    <h3>Participations & Achievements</h3>
    <ul>
        <li>Participated in GDG Hackathon</li>
        <li>Winner in Pivot Hackathon</li>
        <li>Web Development course completion certificate by Udemy</li>
    </ul>
</body>
</html>
```



127.0.0.1:5500/lab.html

RESUME

Introduction

I am Penkey Prajna Sree. I am currently studying III year of B-TECH in Vishnu Institute of Technology, Bhimavaram in the department of Information Technology.I have looking for some internships to have some experiences and gain some skills in python and AI Technologies. Lorem ipsum dolor sit amet consectetur, adipisicing elit. Consequatur dolorum nam est voluptatibus corporis accusamus labore totam ex? Exercitationem, dicta fugit! Perspiciatis nesciunt, fuga perferendis sit minima voluptate inventore incidunt. Lorem ipsum dolor sit amet consectetur adipisicing elit. Numquam quod, cupiditate repudiandae necessitatibus animi suscipit! Laborum ratione aspernatur debitis eum veniam distinctio provident accusamus quod quidem dolore, eos numquam sit? Lorem ipsum dolor sit, amet consectetur adipisicing elit. Iste voluptates debitis a dolorem nihil neque perferendis modi officiis! Aperiam doloremque dolores ducimus explicabo possimus alias aspernatur tenetur praesentium mollitia quam. Lorem ipsum dolor sit amet consectetur adipisicing elit. Inventore necessitatibus natus, recusandae incidunt quasi numquam enim id corrupti iure adipisci. Sed nesciunt eum quod autem, totam debitis at soluta cupiditate? Lorem ipsum dolor sit amet consectetur adipisicing elit. Eligendi quo incidunt deleniti asperiores animi voluptatem ab reprehenderit laborum illum consequuntur excepturi, placeat, impedit rem doloremque quia explicabo. Ullam, excepturi odio! Lorem, ipsum dolor sit amet consectetur adipisicing elit. Eligendi placeat voluptatum, quidem est reiciendis nisi iure sit ut animi ab numquam magni maiores illum, mollitia cumque aperiam ex. Impedit, harum.

Qualifications

Qualification	School/College Name	percentage
B.Tech	Vishnu Institute of Technology	9.46(CGPA)
Intermediate	Sasi Junior College(CBSE)	80%
10th	Sasi English Medium High School	98%

Hobbies

- Playing Badminton
- Listening music
- Listening podcasts

Participations & Achievements

- Participated in GDG Hackathon
- Winner in Pivot Hackathon
- Web Development course completion certificate by Udemy

2. Demonstrate responsive web design using 4X4 Responsive web layout

Web.css

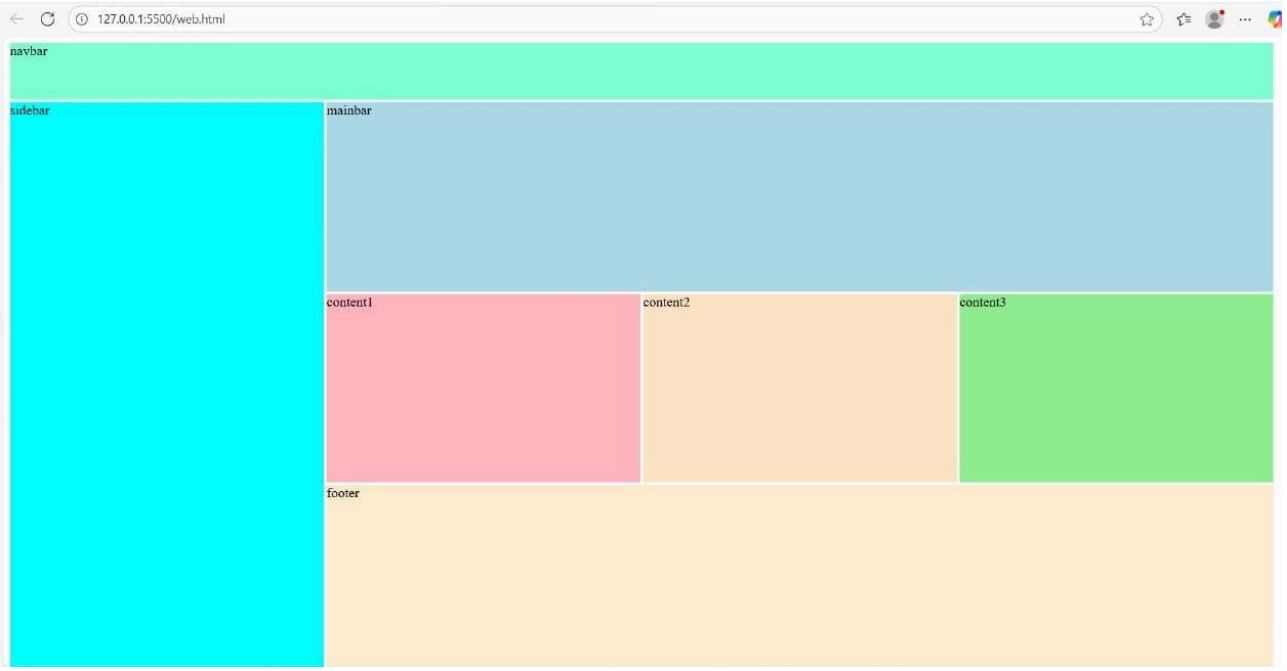
```
.container{
  display:grid;
  height:100vh;
  grid-template-columns:1fr 1fr 1fr 1fr;
  grid-template-rows:0.3fr 1fr 1fr 1fr;
  grid-template-areas:"nav nav nav nav"
    "sidebar main main main"
    "sidebar content1 content2 content3"
    "sidebar footer footer footer";
  grid-gap:0.2rem;
}
nav{
  background-color:aquamarine;
  grid-area:nav;
}
main{
  background-color:lightblue;
  grid-area:main;
}
#sidebar{
  background-color:cyan;
  grid-area:sidebar;
}
#content1{
  background-color:lightpink;
  grid-area:content1;
}
```

```
#content2{
  background-color:bisque;
  grid-area:content2;
}
#content3{
  background-color:lightgreen;
  grid-area:content3;
}
footer{
  background-color:blanchedalmond;
  grid-area:footer;
}
@media only screen and (max-width:550px){
  .container{
    grid-template-columns:1fr;
    grid-template-rows:0.2fr 0.5fr 0.4fr 0.5fr 0.5fr 0.5fr 0.3fr;
    grid-template-areas:"nav"
      "sidebar"
      "main"
      "content1"
      "content2"
      "content3"
      "footer";
  }
}
```

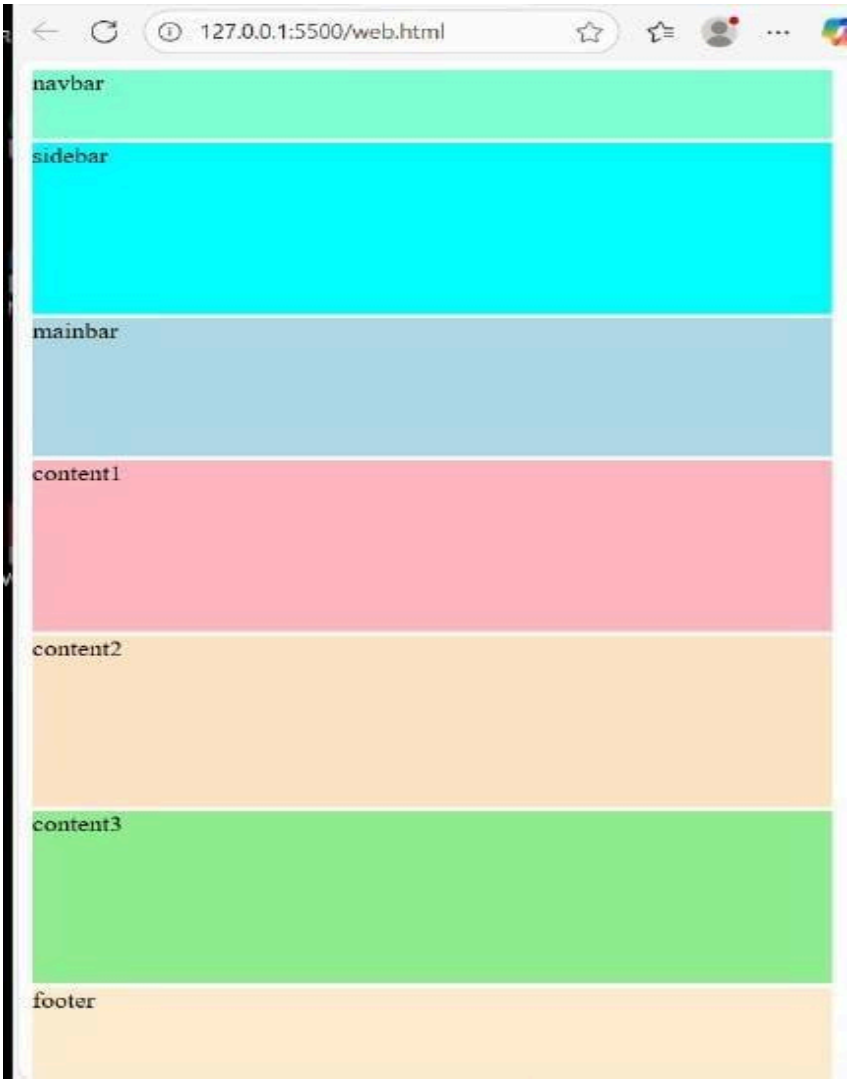
Web.html

```
<!DOCTYPE html>
<html lang="en">
 <head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width,initial-scale=1.0">
   <title>Sample Page</title>
   <link rel="stylesheet" href="sample.css">
  </head>
  <body>
    <div class="container">
      <nav>navbar</nav>
      <main>mainbar</main>
      <div id="sidebar">sidebar</div>
      <div id="content1">content1</div>
      <div id="content2">content2</div>
      <div id="content3">content3</div>
      <footer>footer</footer>
    </div>
  </body>
</html>
```

Desktop Navigation:



Mobile Navigation:



3. Build a web application that can fetch student exam result using NodeJS & Express

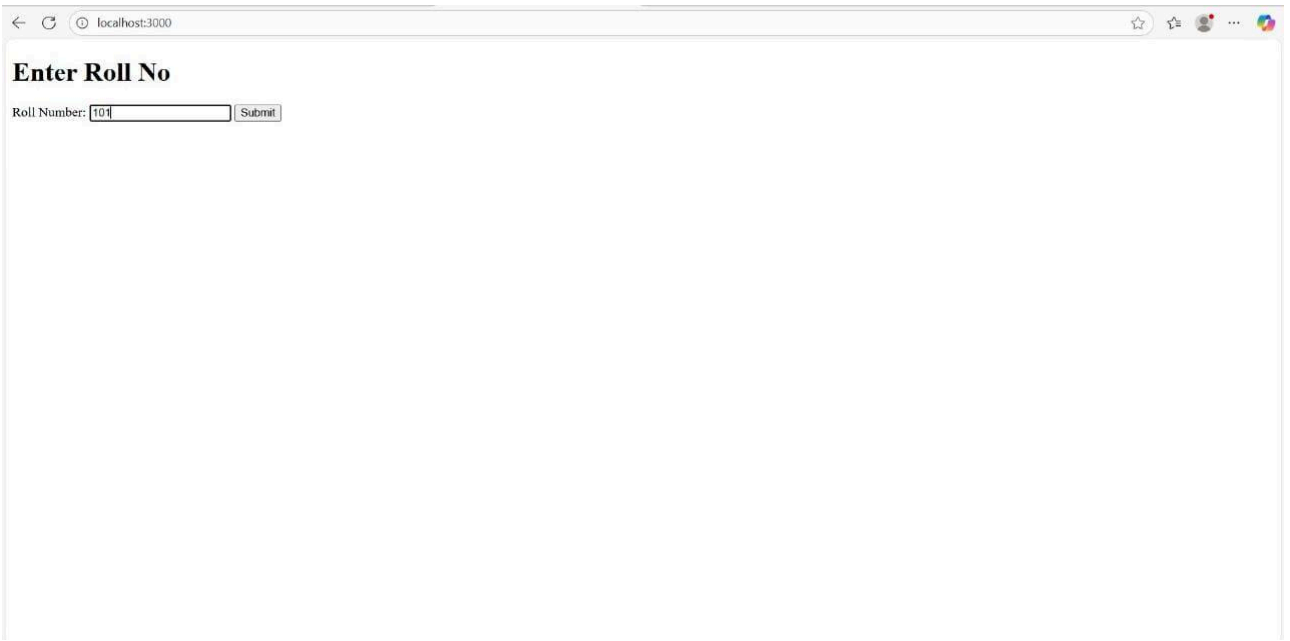
views/home.ejs

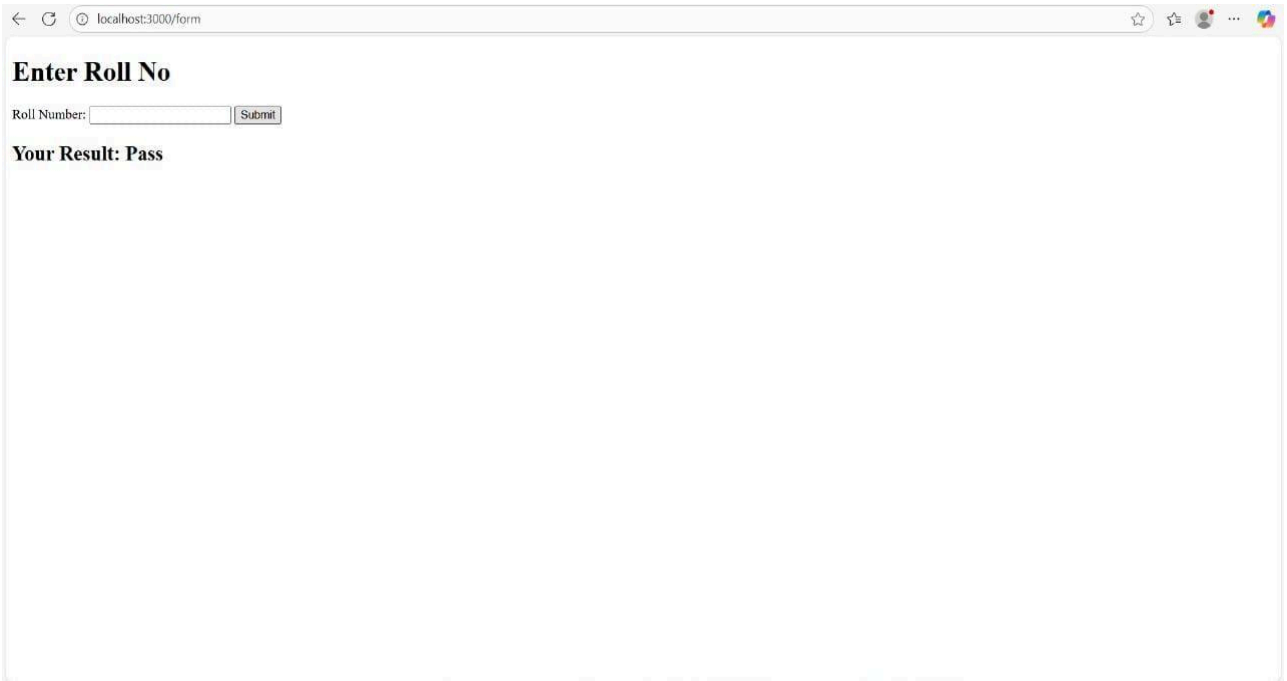
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Roll Number Form</title>
</head>
<body>
    <h1>Enter Roll No</h1>
    <form action="/form" method="POST">
        <label for="rollno">Roll Number:</label>
        <input type="text" name="rollno" id="rollno" required>
        <button type="Submit">Submit</button>
    </form>
    <% if (result) { %>
        <h2>Your Result: <%= result %></h2>
    <% } %>
</body>
</html>
```

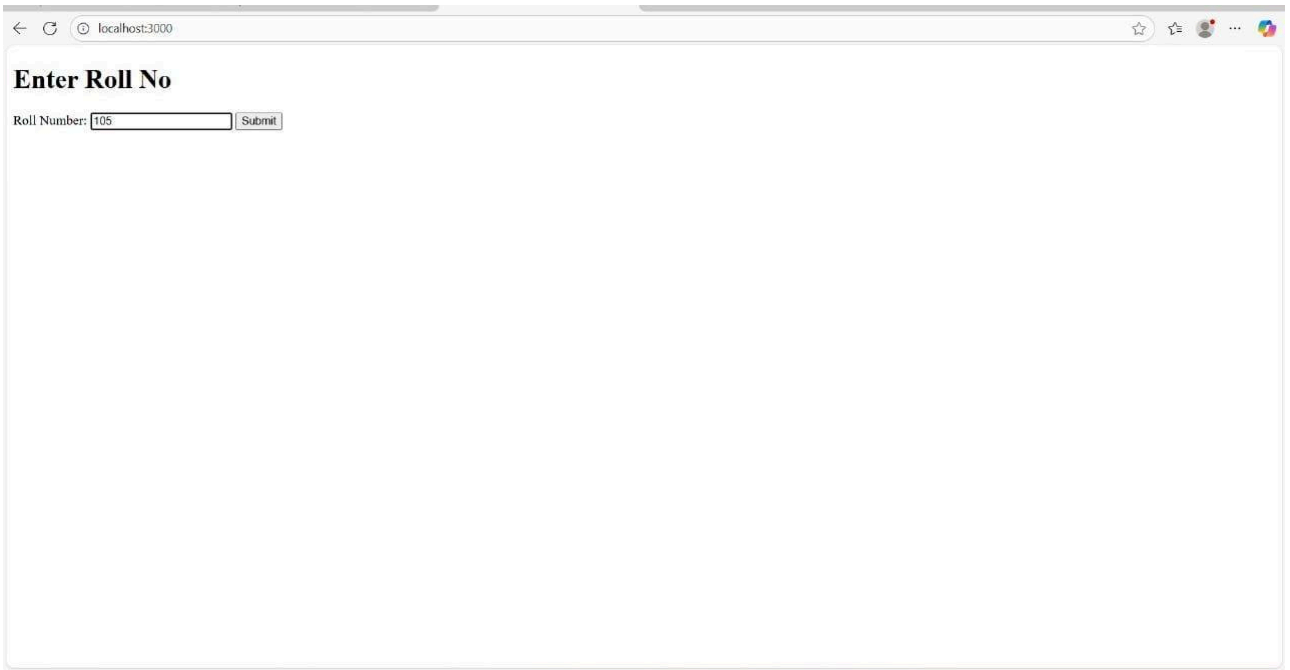
server.js

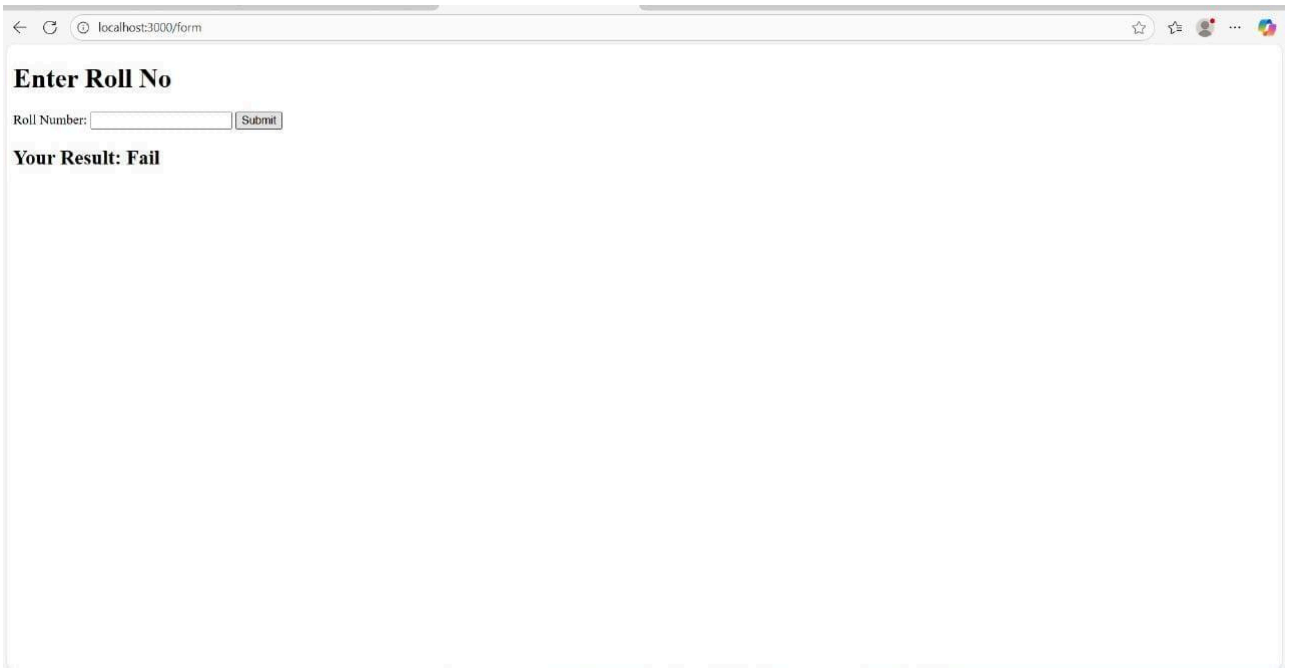
```
const express = require("express");
const bp = require("body-parser");
const app = express();

app.set("view engine", "ejs");
app.use(bp.urlencoded({ extended: true }));

app.get("/", (req, res) => {
    res.render("home", { result: null });
});

app.post("/form", (req, res) => {
    let rollno = req.body.rollno;
    const list = [101, 102, 103, 104];
    let result = "Fail";

    for (let i = 0; i < list.length; i++) {
        if (rollno == list[i]) {
            result = "Pass";
            break;
        }
    }
```

```
    res.render("home", { result: result });
});

app.listen(3000, () => {
    console.log("Server started on port 3000");
});
```









4. Demonstrate various CRUD operations in mongoDB

MongoDB: Inserting a document into a collection (Create)

```
db.student.insert({
        regNo: "3014",
        name: "Test Student",
```

```
        course: {
                courseName: "MCA",
                duration: "3 Years"
        },
        address: {
                city: "Bangalore",
                state: "KA",
                country: "India"
        }
})
```

MongoDB: Querying a document from a collection(Read)

db.student.find()

To find specific value: db.collection_name.find({"fieldname":"value"})

db.students.find({"regNo":"3014"})

MongoDB: Updating a document in a collection(Update)

```
db.student.update({
        "regNo": "3014"
},
$set:
{
        "name":"Viraj"
})
```

MongoDB: Removing an entry from the collection(Delete)

db.collection_name.remove({"fieldname":"value"})

db.collection_name.remove({"regno":"3014"})

5. Build a web application to demonstrate MONGO DB CRUD operations(Write and Read)

Form.ejs

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8" />
 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
 <title>Student Complaint Form</title>
 <link rel="stylesheet" href="/style.css" />
</head>
<body>
 <div class="container">
   <h1>►► Student Complaint Form</h1>

   <form action="/submit" method="POST" class="complaint-form">
     <div class="form-group">
       <label for="name">Full Name</label>
         <input
         type="text"
         id="name"
         name="name"
         placeholder="Enter your full name"
         required
       />
     </div>

     <div class="form-group">
       <label for="roll">Roll Number</label>
       <input
         type="text"
         id="roll"
         name="roll"
         placeholder="Enter your roll number"
         required
       />
     </div>

     <div class="form-group">
       <label for="email">Email</label>
       <input
         type="email"
         id="email"
         name="email"
         placeholder="Enter your email"
         required
       />
```

```
    </div>

    <div class="form-group">
      <label for="branch">Branch</label>
      <select id="branch" name="branch" required>
        <option value="" disabled selected>Select your branch</option>
        <option value="CSE">CSE</option>
        <option value="IT">IT</option>
        <option value="ECE">ECE</option>
      </select>
    </div>

    <div class="form-group">
      <label for="complaintType">Complaint Type</label>
      <select id="complaintType" name="complaintType" required>
        <option value="" disabled selected>Select complaint type</option>
        <option value="Academic">Academic</option>
        <option value="Extra-curricular">Extra-Curricular</option>
        <option value="Travel">Travel</option>
        <option value="Library">Library</option>
      </select>
    </div>

    <div class="form-group">
      <label for="description">Complaint Description</label>
      <textarea
        id="description"
        name="description"
        rows="4"
        placeholder="Describe your issue here..."
        required
      ></textarea>
    </div>

    <button type="submit">Submit Complaint</button>
  </form>
 </div>
</body>
</html>
```

index.ejs

```
const express = require('express');
const bodyParser = require('body-parser');
const { MongoClient } = require('mongodb');
const app = express();
const port = 3900;
```

```
const uri =
"mongodb+srv://suresh:10pa1a1218@cluster0.kmhoe.mongodb.net/?retryWrites=true&w=majority
&appName=Cluster0";
app.set("view engine", "ejs");
app.use(bodyParser.urlencoded({ extended: true }));
let db;
async function connectDB() {
 try {
   const client = new MongoClient(uri);
   await client.connect();
   db = client.db('mydb');
   console.log("Connected to MongoDB");
 } catch (error) {
   console.error("MongoDB connection error:", error);
 }
}
connectDB()
app.get("/", (req, res) => {
 res.render("form");
});
app.post("/submit", async (req, res) => {
 const { name, roll, email, branch, complaintType, description } = req.body;
 console.log("\nComplaint Received:");
 console.log("Name:", name);
 console.log("Roll Number:", roll);
 console.log("Email:", email);
 console.log("Branch:", branch);
 console.log("Complaint Type:", complaintType);
 console.log("Description:", description);
 console.log("------------------------------------");
 try {
   const complaints = db.collection('complaints');
   await complaints.insertOne({
     name : name,
     roll : roll,
     email: email,
     branch: branch,
     complaintType: complaintType,
     description: description,
```

```
   });
   res.send("Complaint submitted successfully!");
  } catch (err) {
   console.error("Error inserting complaint:", err);
   res.status(500).send("Failed to submit complaint");
 }
});
 connectDB().then(()=>{
async function getData(){
 const complaints = db.collection('complaints');
  const data = await complaints.find().toArray();
 console.log(data);
   }
getData();
 app.get('/data', async(req,res)=>{
 const alldata = await getData();
 console.log(alldata);
      })
 })
app.listen(port, () => {
 console.log(`Server running at http://localhost:${port}`);     });
```

6. Build a web application to demonstrate API CALLS using OpenWeather API

index.js

```
const express = require('express');
const axios = require('axios');
const app = express();
const port = 4000;
app.set("view engine", "ejs");
app.use(express.urlencoded({ extended: true }));
app.use(express.static("public"));
app.get("/", (req, res) => {
 res.render("index", { weather: null, error: null });
});
app.post("/", async (req, res) => {
```

```
 const city = req.body.city;
 const apiKey = "18656385803980ca07b53f81f587df49";
 const url =
`https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${apiKey}&units=metric`;
 try {
   const response = await axios.get(url);
   console.log(response);
   const data = response.data;
   const weather = {
     city: data.name,
     temp: data.main.temp,
     description: data.weather[0].description,
     icon: data.weather[0].icon,
   };
   res.render("index", { weather, error: null });
 } catch (error) {
   res.render("index", { weather: null, error: "City not found or invalid input" });
 }
});

app.listen(port, () => console.log(`Server running on http://localhost:${port}`));
```

Form.ejs

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8" />
 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
 <title>Weather App</title>
 <!-- <link rel="stylesheet" href="/styles.css" type="text/css"/> -->
</head>
<body>
 <div class="container">
   <h1>►► Weather App</h1>
   <form action="/" method="POST">
     <input type="text" name="city" placeholder="Enter city name" required />
     <button type="submit">Get Weather</button>
   </form>
```

```
  <% if (weather) { %>
    <div class="weather-box">
      <h2><%= weather.city %></h2>
      <img src="https://openweathermap.org/img/wn/<%= weather.icon %>@2x.png" alt="weather
icon" />
      <p><%= weather.description %></p>
      <h3><%= weather.temp %>°C</h3>
    </div>
  <% } %>

  <% if (error) { %>
    <p class="error"><%= error %></p>
  <% } %>
 </div>
</body>
</html>
```